

Business Intelligence & Operational Glossary

Standardized Enterprise Metrics, Terminology & Formula Specifications

1. Executive Summary & Governance Standards

This Business Intelligence Glossary serves as the single source of truth (SSOT) for analytical terminology, key performance indicators (KPIs), and dimensional attributes across Northwind Traders. All executive dashboards, reporting SQL queries, and data warehouse semantic models must strictly adhere to these standard calculations.

2. Core Financial & Revenue Metrics

2.1 Revenue (Net Sales Revenue)

Definition: Total monetary value realized from fulfilled sales of goods to customers after deducting line-item volume discounts.

Formula: $\text{Net Revenue} = \sum (\text{Order Details.UnitPrice} \times \text{Order Details.Quantity} \times (1 - \text{Order Details.Discount}))$

Exclusions: Excludes canceled orders, pending holds, freight charges, and refunded damaged goods.

2.2 Profit (Net Gross Profit)

Definition: Financial gain calculated by subtracting the direct wholesale procurement Cost of Goods Sold (COGS) from Net Revenue.

Formula: $\text{Gross Profit} = \text{Net Revenue} - \text{Cost of Goods Sold (COGS)}$

Application: Used to assess product line profitability and regional margin contribution.

2.3 Gross Margin Percentage

Definition: Proportion of revenue retained as profit after incurring direct inventory procurement costs, expressed as a percentage.

Formula: $\text{Gross Margin \%} = (\text{Gross Profit} / \text{Net Revenue}) \times 100$

2.4 Order Value (Total Transaction Value)

Definition: Total monetary bill for a single completed sales order header, combining net line-item sales product values with shipping freight charges.

Formula: $\text{Order Value} = \text{Net Revenue_Order} + \text{Orders.Freight}$

2.5 Average Order Size (AOS) / Average Order Value (AOV)

Definition: Average net monetary revenue generated per fulfilled order over a specific reporting timeframe.

Formula: $\text{AOS} = \text{Total Net Revenue} / \text{Total Count of Fulfilled Orders}$

3. Supply Chain & Operational Metrics

3.1 Inventory Turnover Ratio

Definition: Operational ratio measuring how many times total inventory is sold and replaced over a given annual or quarterly period.

Formula: $\text{Inventory Turnover} = \text{Annual COGS} / \text{Average Inventory Value}$

Target: High turnover indicates efficient inventory management and strong demand.

3.2 Lead Time

Definition: Total elapsed calendar time (in days) from the exact moment a stock replenishment Purchase Order is transmitted to a supplier until stock is delivered, inspected, and logged into warehouse inventory.

Formula: $\text{Lead Time} = \text{Received Date} - \text{Purchase Order Sent Date}$

3.3 Days Sales Outstanding (DSO)

Definition: Measure of the average number of days it takes for Northwind to collect payment from wholesale trade credit customers after sales invoice issuance.

Formula: $\text{DSO} = (\text{Accounts Receivable Balance} / \text{Total Credit Sales}) \times \text{Total Days in Period}$

4. Relationship & Operational Entities

4.1 Supplier

Definition: External commercial vendor or manufacturer contracted by Northwind Traders to supply wholesale specialty foods, beverages, or raw produce (indexed in database via SupplierID).

4.2 Customer Lifetime Value (CLV / LTV)

Definition: Prediction of cumulative net gross profit contributed by a customer account over the entire duration of its business relationship with Northwind Traders.

Formula: $\text{CLV} = \text{Average Annual Gross Profit Contribution per Customer} \times \text{Average Customer Relationship Lifespan (Years)}$

4.3 Category

Definition: High-level product taxonomy classification used to organize product catalog items into logical groups (e.g., Beverages, Dairy Products, Seafood) for market performance benchmarking.

4.4 Territory

Definition: Designated geographic commercial boundaries assigned to specific sales representatives to direct market coverage and evaluate sales quota achievements.

4.5 Employee

Definition: Internal Northwind personnel record identified in the enterprise system by EmployeeID, assigned to organizational departments, reporting hierarchies, and regional territories.

5. Metric Summary Matrix

Metric Name	Granularity	Primary Business Owner	Target KPI Benchmark
Net Revenue	Line Item / Order / Regional	Finance & Sales Leadership	+12% YoY Growth
Gross Margin %	Product / Category	Procurement & Finance	≥ 35.0% Overall
Average Order Size	Customer Account	VP of Sales	> \$1,250 per order
Inventory Turnover	Product SKU / Warehouse	Logistics & Warehousing	≥ 8.0x per annum
Supplier Lead Time	Supplier ID	Supply Chain Operations	≤ 14 Days Average
Days Sales Outstanding	Accounts Receivable	Credit & Collections	≤ 38 Days